1.1 Which line coding scheme has synchronization issues. Ans. (b)
a) AMI with B8ZS
b) NRZ - I
c) Manchester
d) All of these
e) None of the above

1.2 In the design of a transmission channel, we would like to use one pulse to carry 4 bits of information. What will be the maximum E_s/N_0 (in dB) to ensure reliable transmission. Ans a
a) 24.7 dB
b) 11.7 dB
c) 4.7 dB

$2B \underbrace{\log_2 M}_{4} < B \underbrace{\log_2}(1+SNR)$

d) It depends on the bandwidth of the transmission channel.
e) None of the above

1.3 Assume host A wants to send a large file of 2 MB to host B. The path from host A to host B are directly connected by a single link of speed 1 Gbps. Assume the link between A and B are 1 Km long and the propagation speed in the link is 2*10^8 meters/sec. How long does it take to transmit the whole message from host A to host B?
(Given: 1 byte = 8 bits, 1 Mbps = 10^6 bits/second, 1 KB = 10^3 bytes).
a) 2 ms
b) 8 ms
c) 16 ms   e
d) 17 ms
e) None of the above

16 Mbits

$\frac{16 \times 10^6}{1 \times 10^9} = 16 \times 10^{-3}$

$\frac{1km}{2 \times 10^8} = 5 \times 10^{-6} < 1ms$

1.4 Which cannot be done at data link layer?
a) Addressing
b) Congestion Control   b
c) Error Control
d) Flow Control
e) None of the above

1.5 Which kind of delays can be found from end to end in a packet switched network?
a) Processing Delay   e
b) Transmission Delay
c) Propagation Delay
d) Queuing Delay
e) All of the above

Question 2 (10 Marks)

Compute the checksum using the binary approach or modulo arithmetic approach for the following three 16-bit words 0011100111110011, 1111000011110000, 1010101010101010. The source device sends the data and checksum to the destination device across the network. Assume that the destination device receives the data and checksum correctly without error. Show, by means of computations, how the destination device verifies that there is no error.

Question 3
Suppose a transmitter transmits a 16-bit sequence as below using different line coding techniques.
11111111 00000000

a) If NRZ-I is used as the coding technique, where a binary 0 results in no change in polarity, and a binary 1 results in a change in polarity, plot the transmitted waveform. You may assume the first bit 1 is encoded at positive level at +V.

b) If AMI together with B8ZS encoding rule is used as the coding technique, plot the transmitted waveform. You may assume the first bit 1 is encoded at positive level at +V. The replacement rule for B8ZS is substitute 8 consecutive 0s with 000VB0VB.